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Unit – 1 : Session – 4 : SLO – 2 | Codes and parity

■ 5 MCQs

1. b — 1110 (Gray of 1011: $1, 1^0=1, 0^1=1, 1^1=0$)
2. c — 1010 ($7 + 3 = 10$)
3. a — 0000 (0) with carry ($9+3=12 \rightarrow 1001+0011=1100, +0110 \rightarrow 0010$ carry 1... i.e., BCD 12; as single digit it is 0010 with carry)
4. a — 65
5. b — Ensures total number of 1s is even

■ 2 Two-Mark Questions

1. Gray 10011 to binary: $b_4 = g_4 = 1; b_3 = b_4 \oplus g_3 = 1 \oplus 0 = 1; b_2 = b_3 \oplus g_2 = 1 \oplus 0 = 1; b_1 = b_2 \oplus g_1 = 1 \oplus 1 = 0; b_0 = b_1 \oplus g_0 = 0 \oplus 1 = 1$. Binary = 11011.
2. Decimal 2: BCD = 0010; Excess-3 = 0010 + 0011 = 0101.

■ 1 Scenario-Based Long Answer

- Pipeline: take decimal digit 5 $\rightarrow$ BCD 0101 $\rightarrow$ Excess-3 1000 $\rightarrow$ count 1s in 1000 = 1 (odd) $\rightarrow$ even-parity bit = 1 $\rightarrow$ transmitted frame 1 1000.
- At the receiver, the 1s in the 5-bit frame are counted: 11000 has two 1s (even) $\rightarrow$ no error. If a bit flips in transit (e.g. 11010, three 1s), the parity check fails and the digit is rejected/retransmitted. BCD $\rightarrow$ XS-3 $\rightarrow$ parity thus gives both decimal-friendly encoding and single-bit error protection.

■ 5 Match the Following

- Binary to Gray — a) Copy first bit, then XOR
- ASCII of 'B' — b) 66
- BCD Addition — c) Add 0110 if result > 1001
- XS-3 code for 2 — d) 0101
- Even Parity — e) Total number of 1s is even

■ 5 Fill in the Blanks

1. 10101 (Gray of 11001: $1, 1^1=0, 1^0=1, 0^0=0, 0^1=1$)
2. B 3. 0110 4. 0110 ($3+3=6$) 5. error

■ 1 Scenario-Based Small Activity

- Data 1100110: count of 1s = 4 (already even) $\rightarrow$ even parity bit = 0.
- Appended as MSB, final 8-bit result: 01100110.

■ 5 True or False

1. FALSE — Standard ASCII uses 7 bits.
2. TRUE
3. TRUE

4. FALSE — Excess-3 is non-weighted. 5. FALSE — Even parity ensures an even number of 1s.